

Infosys Springboard Virtual Internship 7.0

MarketMind AI

AI-Powered Small Business Sales Intelligence Platform

Team Contribution Report

Milestone-wise Individual Contributions (Milestones 1-4)

Team Members

1. Divyanka Singh
2. Garvit Khuteta
3. G. Vishnuvardhan Rao
4. Akshaya Bale
5. Komal Kumari
6. Tejananda

1. Milestone 1 - Project Foundation, Roles & Setup

Divyanka Singh — Frontend UI & Integration

- Set up the React + Vite frontend environment and developed the dashboard UI for all four user roles (Business Owner, Store Manager, Sales Executive, System Administrator).
- Improved login form validation and handling of invalid or incomplete input.
- Replaced the mock login with real backend authentication; implemented logout, automatic token refresh, and persistent login after page refresh.
- Fetched the logged-in user's profile and role from the backend and ensured the correct role-based dashboard opens, with permissions always controlled by the backend for better security.

Garvit Khuteta — Backend Development & Architecture

- Set up the backend environment and developed the APIs required for communication between the frontend and the database.
- Structured the backend to support authentication and future AI features, creating a strong foundation for integrating all project components.
- Built an asynchronous REST API framework using Fast API, SQL Alchemy 2.0, Pydantic v2, and Alembic migrations.
- Configured dual database compatibility for local development (SQLite) and production cloud deployment (Neon PostgreSQL).
- Implemented 4-tier Role-Based Access Control (Business Owner, Store Manager, Sales Executive, System Administrator) with Argon2id password hashing and rotating JWT tokens.
- Created automated CLI data-seeding scripts (seed_demo.py, bootstrap.py) to provision initial evaluation data and demo accounts.

G. Vishnuvardhan Rao — Frontend Development & Admin Dashboard Integration

- Worked on frontend development across Milestones 1 and 2.
- Connected the Admin dashboard with backend features such as the user list, user invitations, role management, account enable/disable functionality, and audit logs.
- Added handling for loading, invalid login, permission-denied, empty-data, and server-error states to improve the user experience.
- Configured the API base URL using environment variables and verified that login, logout, session persistence, role-based dashboard access, and backend data integration worked correctly, along with the project's lint and build checks.

Akshaya Bale — System Design, Requirements & Documentation

- Prepared and documented the project statement and problem definition, covering sales intelligence, customer behavior analysis, business analytics, inventory intelligence, forecasting, and AI-driven decision support.
- Defined and documented the major project objectives and mapped them to planned business-intelligence and AI capabilities.
- Defined the four major user roles (Business Owner, Store Manager, Sales Executive, Administrator) and documented their responsibilities and access areas.

- Designed and documented the system architecture, covering the React frontend, FastAPI backend, REST APIs, database, authentication, RBAC, data preprocessing, ML modules, analytics, forecasting, and reporting.
- Documented the conceptual database schema and the data flow from datasets/processing to backend services and application-level consumption.
- Prepared and organized the complete Milestone 1 documentation package (overview, problem statement, objectives, user roles, team responsibilities, workflow, architecture, database design, deliverables, and outcomes) and managed related repository cleanup on GitHub.

Komal Kumari — Dataset Preparation

- Collected and analyzed the sales dataset and studied its attributes.
- Performed data cleaning by handling missing values, duplicates, and inconsistencies.
- Preprocessed the data so it was ready for analysis and future AI model development, ensuring a clean and reliable dataset for upcoming milestones.

Tejananda — Frontend Development

- Set up the React and Vite frontend environment.
- Developed the frontend structure, including the dashboards for different user roles.
- Organized the project into reusable components so it could be easily maintained and integrated with the backend, providing the basic frontend framework for the application.

2. Milestone 2 - Customer Intelligence, Forecasting & Reports

Divyanka Singh — AI Forecasting Dashboard — Frontend

- Developed the AI Forecasting dashboard frontend for the Business Owner and Store Manager roles, organizing forecasting information into clear dashboard sections while maintaining the existing UI style.

Garvit Khuteta — Customer Segmentation & Workspace/Billing Backend

- Worked with Komal Kumari on customer segmentation and behavioral analysis using customer-level purchasing features such as recency, frequency, revenue, order value, product variety, return rate, and engagement.

- Contributed backend support for role-based workspaces and dashboards (Business Owner, Store Manager, Sales Executive) and the GST-compliant Indian B2B billing and invoicing features, including the credit receivables aging engine.
- Carried out the final integration of the customer analytics and segmentation work into the application.

G. Vishnuvardhan Rao — Frontend Development (continued)

- Continued frontend development work spanning Milestones 1 and 2, including contributions to Admin dashboard integration and frontend testing described under Milestone 1.

Akshaya Bale — Customer Intelligence, Forecasting UI & Reports Module

- Analyzed customer purchasing behavior using RFM (Recency, Frequency, Monetary) features together with additional engagement indicators (active months, product variety, repeat purchasing, returns).
- Worked on the business interpretation of the generated customer segments (Champions, Needs Attention, Hibernating) for reporting; the underlying clustering model itself was team-owned.
- Designed the Milestone 2 AI forecasting UI for Revenue Forecast (Business Owner), Personal Sales Forecast (Sales Executive), and Product Demand Forecast (Store Manager), covering 7/14/30-day forecast horizons.
- Converted the placeholder ReportsModule.jsx into a functional analytics and forecasting reporting workspace with role-aware report/forecast selection, model information, and forecast insights.
- Integrated the reporting frontend with the authenticated Fast API backend, consuming customer, customer-segment, and forecasting (revenue/personal/demand/monitoring) APIs.
- Added CSV export and browser-based print/PDF export functionality, and handled loading, empty-data, and API-failure states in the reporting UI.
- Debugged local frontend/backend integration issues (Vite server setup, CORS, API routing, authentication verification) to ensure reliable report data loading.

Komal Kumari — Customer Analytics

- Collected the customer dataset and analyzed its attributes to identify features useful for understanding customer purchasing behavior.
- Worked with Garvit Khuteta on customer segmentation and behavioral analysis using recency, frequency, revenue, order value, product variety, return rate, and engagement.

- Built the CustomerAnalyticsReport.jsx component to present customer analytics through the application.
- Worked with Akshaya Bale on the DemandForecastReport.jsx component for displaying demand-forecasting information.
- Final integration of this milestone's analytics work was carried out by Garvit Khuteta.

3. Milestone 3 - AI Recommendations, Churn & Anomaly Detection

Divyanka Singh — Intelligent Product Recommendation System (Full-stack / ML)

- Developed the recommendation-system frontend module and the corresponding backend functionality and integrated the frontend with the backend API/service.
- Implemented Collaborative Filtering and Association Rule Mining to identify product relationships from transaction patterns.
- Implemented Precision@K and Recall@K evaluation metrics to assess the relevance and coverage of top K recommendations.

Garvit Khuteta — AI Inference Engines & Business Copilot

- Built custom, role-tailored dashboard UI views for the Business Owner, Store Manager, Sales Executive, and Administrator roles.
- Implemented granular Role-Based Access Control (RBAC) with strict feature- and route-level gating, so each user only accesses their assigned actions.
- Built a 2-step Purchase Order workflow (Manager drafts → Owner approves) with automated email notification and CSV export.
- Implemented GST-compliant A5-printable tax billing with CGST/SGST/IGST breakdown and automated number-to-words currency conversion.
- Built in the B2B credit ledger system, including customer credit limits, Net 30/60 payment terms, and quick-settlement functionality.
- Built in the 60-day staff audit system, covering employee activity logs, token-based invitations, and sales target tracking.
- Integrated the AI Copilot as both a dedicated full-screen chat interface and a floating in-app assistant.
- Built the product recommendation engine using Collaborative Filtering for cross-sell bundle suggestions.

- Built churn prediction using RFM scoring with 1-click WhatsApp/Email outreach triggers.
- Built the anomaly detection guard using Isolation Forest to flag revenue dips and data leaks.
- Built demand forecasting for 7/14/30-day SKU-level predictions using ARIMA/Prophet models.
- Built the model-training pipeline, including a background training model and live status refresh.

G. Vishnuvardhan Rao — Churn Prediction, Sales Activity & Anomaly Detection

- Worked on churn prediction, sales activity analysis, and anomaly detection for this milestone.

Akshaya Bale — Personalized Business Recommendations & Alerts

- Implemented collaborative filtering and recommendation techniques to identify relevant product recommendations from customer purchasing/transaction patterns.
- Worked on generating personalized product recommendations based on customer behavior and available recommendation insights.
- Converted recommendation outputs into actionable, business-facing recommendations that support cross-selling and customer engagement decisions.
- Worked on generating and presenting personalized business alerts based on relevant analytical and AI-driven conditions, focusing on making them understandable and useful for business users.
- Integrated recommendation and alert outputs into the existing role-based application and dashboard workflow, maintaining consistency with the roles defined in Milestone 1.
- Converted technical analytics and AI outputs into business-facing insights to support decision-making, rather than presenting only raw model results.
- Validated recommendations and alert presentation across loading, empty-result, unavailable data, and error scenarios.

Komal Kumari —just revise the work completed through milestone 1 & 2.

4. Milestone 4 - Testing, Security, Deployment & Documentation

Divyanka Singh — Responsive UI & Project Documentation

- Checked and improved the responsiveness of dashboard and application layouts across different screen sizes, adjusting behavior where layout issues were identified.
- Prepared and organized the project's main documentation, working alongside G. Vishnuvardhan Rao.

Garvit Khuteta — Security, Governance, Testing & Deployment

- - Enforced strict multi-tenant data isolation so no business could access another's data.
 - Enforced RBAC-based feature restrictions across both backend and frontend layers.
 - Implemented 6-digit email OTP verification (via Resend/SMTP) for signup and password resets.
 - Built a tiered token-bucket rate limiter to defend against API spam and brute-force attempts.
 - Implemented safe file-upload validation using magic-byte checks and a 2MB size cap for brand assets.
 - Implemented global exception handlers to sanitize errors and mask internal database schemas.
 - Fixed a `matches Payment` reference error crashing the Sales tab (`SalesModule.jsx`).
 - Fixed a missing `Send` icon import crashing the Purchase Order modal (`ManagerDashboard.jsx`).
 - Fixed an undefined `category Distribution` reference crashing Owner login.
 - Fixed an `onReissueToken` reference error in the Team Management modal.
 - Fixed demo data leakage where new/blank accounts inherited mock store metrics.
 - Fixed an avatar upload bug where `resolveApiAsset` incorrectly prepended URLs to blob/data preview links.
 - Fixed a role-permission bug by blocking the business-setup modal for Store Managers and routing them to inventory instead.
 - Fixed a B2B search bug, restore dropped search results and add query support by Name, Email, Phone, and GSTIN.
 - Fixed multi-store reactivity issues by correcting `use Memo` inventory dependencies for multi-branch filtering.
 - Improved frontend resilience with Error Boundary and null-safety guards across navigation and routing.
 - Fixed AI tenant-isolation leaks where Churn, Forecast, and Anomaly models shared data across tenants.
 - Corrected credit-aging calculations for overdue receivables (7, 15, and 30+ day buckets).

- Fixed a currency-display bug by removing hardcoded \$ symbols in favor of consistent INR (₹) formatting.
- Added an authentication fallback with autoactivation when the email/SMTP server is unconfigured.
- Added Hindi translation support with 250+ bilingual dictionary keys across the entire UI.
- Applied Indian "Vyapar" branding, including INR (₹) formatting, the Dukaan logo, and Outfit fonts.
- Refined the landing page with an aligned navbar, isolated FAQ section, and improved responsive design.

G. Vishnuvardhan Rao — Project Documentation

- Worked on the project documentation for this milestone together with Divyanka Singh.

Akshaya Bale — Testing, Validation, Error Handling & Quality Assurance

- Ran and debugged the backend test suite, diagnosing a failure caused by stale hardcoded dates and fixing it by using current UTC dates with relative offsets, bringing the backend suite to a full pass.
- Added test coverage for the forecasting module (unsupported-horizon validation), anomaly detection (severity filtering), and the recommendation module (generation, customer-specific recommendations, category affinity, customer tier, inventory factors, analytics, and empty-result scenarios).
- Introduced frontend automated testing from scratch using Vitest, React Testing Library, jest-dom, and jsdom, writing tests for the API client, recommendation service, anomaly service, and the Reports, Anomaly Detection, and Product Recommendations UIs.
- Debugged frontend test fixtures (e.g., a missing potential_revenue_boost field) without changing production code.
- Validated UI error-handling and retry behavior for recommendation and reporting API failures, and for anomaly severity filtering.
- Manually tested the full application workflow end-to-end — business setup, dashboard, AI recommendations, anomaly alerts, reports and forecasts, CSV export, and the train-and-refresh workflow — identifying and resolving a model-state prerequisite issue.
- Delivered the final project-wide testing results: Frontend 19/19, Backend 41/41, Preprocessing 10/10 (Total 70/70 tests passing), 0 frontend lint errors, and a successful production build.

Komal Kumari — Application Testing & Documentation

- Manually tested the full application workflow end-to-end — business setup, dashboard, AI recommendations, anomaly alerts, reports and forecasts, CSV export, and the train-and-refresh workflow — identifying and resolving a model-state prerequisite issue.
- Made a complete document on team's milestone working.